Higher National Diploma in Information Technology
Second Year, Second Semester Examination – 2023
HNDIT4022 - Software Quality Assurance

Instructions for candidates:	No. of questions	: 06
Answer **five** questions **only.**	No. of pages	: 02
All questions carry **equal** marks.	Time	:03 Hours

Question 01 **[Total 20 Marks]**

i. Define the term "Software Quality". (04 Marks)
ii. Briefly explain why testing is necessary for a software project. (04 Marks)
iii. Mention three (03) views of quality. (03 Marks)
iv. What do you mean by cost of quality? (03 Marks)
v. Briefly explain the types of costs included in cost of quality. (06 Marks)

Question 02 **[Total 20 Marks]**

i. What do mean by a "Software review"? (03 Marks)
ii. Briefly explain the difference between formal review and informal review. (04 Marks)
iii. Briefly describe software peer reviews and mention two (02) types of software peer reviews. (04 Marks)
iv. What is software verification? (03 Marks)
v. Briefly explain two (02) types of software verification. (06 Marks)

Question 03 **[Total 20 Marks]**

i. What is static testing? (04 Marks)
ii. Explain the differences between static testing and dynamic testing. (04 Marks)
iii. List four (04) benefits of static testing. (04 Marks)
iv. Mention four (04) manual methods of static testing. (04 Marks)
v. What types of issues can be identified through static code analysis? (04 Marks)

Question 04 **[Total 20 Marks]**

i. Mention two (02) techniques that can be used in each of the following testing types.
   a. Black box testing (02 Marks)
   b. White box resting (02 Marks)
   c. Integration testing (02 Marks)

ii. Briefly explain the two (02) advantages of black box testing. (02 Marks)
iii. List four (04) objectives of system integration testing. (04 Marks)
iv. What is the purpose of Apache JMeter? (04 Marks)
v. What types of testing can Apache JMeter perform? (04 Marks)

Question 05 **[Total 20 Marks]**

i. What is system testing? (02 Marks)
ii. Who performs system testing? (02 Marks)
iii. List four (04) disadvantages of system testing. (04 Marks)
iv. Briefly explain any three (03) types of system testing. (06 Marks)
v. You're testing an e-commerce website. What would you include in a system test? (06 Marks)

Question 06 **[Total 20 Marks]**

i. What is a test environment? (02 Marks)
ii. List any three (03) types of test environments. (03 Marks)
iii. Briefly describe any two (02) major roles included in a software testing team. (04 Marks)
iv. Briefly describe three (03) major tools in Selenium. (03 Marks)
v. Identify selenium testrunner common component and tasks. (08 marks)